

Title

University of Glasgow



University
of Glasgow

Joseph Henry Briggs

Day Month Year

Contents

1	Gravitational Wave Interferometers	6
1.1	Introduction	6
1.2	Measuring Gravitational Waves	6
1.3	Sources of noise when detecting gravitational waves	7
1.4	The future of gravitational wave detectors	7
1.5	conclusion	8

Declaration

Acknowledgements

Abstract

Chapter 1

Gravitational Wave Interferometers

1.1 Introduction

The aim of this chapter is to outline some things about our universe we would like to know, how a gravitation wave allows this to be known, how an interferometer can be used to measure gravitational waves, the fundamental noise sources these detectors face. These noise sources can be reduced by making changes to the readout of the LIGO detector and by building new interferometers that use new technology. In this thesis, power of the laser and BHD is explored. Third generation detectors may operate at 2um, and so 2um detectors were characterised.

1.2 Measuring Gravitational Waves

general relativity (L shaped IFO)

sources of GW

observation runs and discoveries (detector hierarchy)

The need for increasing detections...

1.3 Sources of noise when detecting gravitational waves

shot noise and laser power

quantum efficiency

power recycling

signal recycling to shape quantum noise

squeezed light

frequency dependent squeezing

radiation pressure noise?

other topologies of detectors

other noise sources

1.4 The future of gravitational wave detectors

cyro

silicon

2um or 1550

1.5 conclusion



Bibliography

- [1] B. Willke, P. King, R. Savage, and P. Fritschel. Pre-stabilized laser design requirements. DCC LIGO-T050036-v4, July 2011.
- [2] Kieran Craig, Jessica Steinlechner, Peter G. Murray, Angus S. Bell, Ross Birney, Karen Haughian, Jim Hough, Ian MacLaren, Steve Penn, Stuart Reid, Raymond Robie, Sheila Rowan, and Iain W. Martin. Mirror coating solution for the cryogenic einstein telescope. *Physical Review Letters*, 122(23), jun 2019.
- [3] E. A. Shcherbakov, V. V. Fomin, A. A. Abramov, A. A. Ferin, D. V. Mochalov, and V. P. Gapontsev. Industrial grade 100 kW power CW fiber laser. In *Advanced Solid-State Lasers Congress*. OSA, 2013.
- [4] Junsu Lee, Kwang Hyun Lee, Hwanseong Jeong, Minkyu Park, Ji Hoon Seung, and Jung Hwan Lee. 205 kW all-fiber high-beam-quality fiber amplifier with stimulated brillouin scattering suppression incorporating a narrow-linewidth fiber-bragg-grating-stabilized laser diode seed source. *Applied Optics*, 58(23):6251, aug 2019.
- [5] Yusheng Huang, Ping Yan, Zehui Wang, Jiading Tian, Dan Li, Qirong Xiao, and Mali Gong. 219 kW narrow linewidth FBG-based MOPA configuration

fiber laser. *Optics Express*, 27(3):3136, jan 2019.

- [6] Pengfei Ma, Rumao Tao, Rongtao Su, Xiaolin Wang, Pu Zhou, and Zejin Liu. 189 kW all-fiberized and polarization-maintained amplifiers with narrow linewidth and near-diffraction-limited beam quality. *Optics Express*, 24(4):4187, feb 2016.
- [7] Craig Robin, Iyad Dajani, and Benjamin Pulford. Modal instability-suppressing, single-frequency photonic crystal fiber amplifier with 811 w output power. *Optics Letters*, 39(3):666, jan 2014.
- [8] Clément Dixneuf, Germain Guiraud, Yves-Vincent Bardin, Quentin Rosa, Mathieu Goeppner, Adèle Hilico, Christophe Pierre, Johan Boullet, Nicholas Traynor, and Giorgio Santarelli. Ultra-low intensity noise, all fiber 365 w linearly polarized single frequency laser at 1064 nm. *Optics Express*, 28(8):10960, mar 2020.
- [9] L. Winkelmann, O. Puncken, R. Kluzik, C. Veltkamp, P. Kwee, J. Poeld, C. Bogan, B. Willke, M. Frede, J. Neumann, P. Wessels, and D. Kracht. Injection-locked single-frequency laser with an output power of 220 w. *Applied Physics B*, 102(3):529–538, feb 2011.
- [10] Felix Wellmann, Michael Steinke, Fabian Meylahn, Nina Bode, Benno Willke, Ludger Overmeyer, Jörg Neumann, and Dietmar Kracht. High power, single-frequency, monolithic fiber amplifier for the next generation of gravitational wave detectors. *Optics Express*, 27(20):28523, sep 2019.

- [11] Aaron Buikema, Franklin Jose, Steven J. Augst, Peter Fritschel, and Nergis Mavalvala. Narrow-linewidth fiber amplifier for gravitational-wave detectors. *Optics Letters*, 44(15):3833, jul 2019.
- [12] Nina Bode, Fabian Meylahn, and Benno Willke. Sequential high power laser amplifiers for gravitational wave detection. *Optics Express*, 28(20):29469, sep 2020.
- [13] Fabian Thies, Nina Bode, Patrick Oppermann, Maik Frede, Bastian Schulz, and Benno Willke. Nd:YVO4 high-power master oscillator power amplifier laser system for second-generation gravitational wave detectors. *Optics Letters*, 44(3):719, jan 2019.
- [14] P. Kwee, C. Bogan, K. Danzmann, M. Frede, H. Kim, P. King, J. Pödl, O. Puncken, R. L. Savage, F. Seifert, P. Wessels, L. Winkelmann, and B. Willke. Stabilized high-power laser system for the gravitational wave detector advanced LIGO. *Optics Express*, 20(10):10617, apr 2012.
- [15] the LIGO Scientific Collaboration. Instrument science white paper 2019. LIGO internal document LIGO-T1900409–v3.
- [16] Andrew P. Spencer. *Advanced Techniques in Laser Interferometry for Current and Future Gravitational Wave Detectors*. PhD thesis, University of Glasgow, August 2019.
- [17] A. Buikema, C. Cahillane, G.L. Mansell, C.D. Blair, R. Abbott, C. Adams, R.X. Adhikari, A. Ananyeva, S. Appert, K. Arai, J.S. Areeda, Y. Asali, S.M. Aston, C. Austin, A.M. Baer, M. Ball, S.W. Ballmer, S. Banagiri,

D. Barker, L. Barsotti, J. Bartlett, B.K. Berger, J. Betzwieser, D. Bhattacharjee, G. Billingsley, S. Biscans, R. M. Blair, N. Bode, P. Booker, R. Bork, A. Bramley, A. F. Brooks, D. D. Brown, K. C. Cannon, X. Chen, A. A. Ciobanu, F. Clara, S. J. Cooper, K. R. Corley, S. T. Countryman, P. B. Covas, D. C. Coyne, L. E. H. Datrier, D. Davis, C. Di Fronzo, K. L. Dooley, J. C. Driggers, P. Dupej, S. E. Dwyer, A. Effler, T. Etzel, M. Evans, T. M. Evans, J. Feicht, A. Fernandez-Galiana, P. Fritschel, V. V. Frolov, P. Fulda, M. Fyffe, J. A. Gaiame, K. D. Giardina, P. Godwin, E. Goetz, S. Gras, C. Gray, R. Gray, A. C. Green, E. K. Gustafson, R. Gustafson, J. Hanks, J. Hanson, T. Hardwick, R. K. Hasskew, M. C. Heintze, A. F. Helmling-Cornell, N. A. Holland, J. D. Jones, S. Kandhasamy, S. Karki, M. Kasprzack, K. Kawabe, N. Kijbunchoo, P. J. King, J. S. Kissel, Rahul Kumar, M. Landry, B. B. Lane, B. Lantz, M. Laxen, Y. K. Lecoeuche, J. Leviton, J. Liu, M. Lormand, A. P. Lundgren, R. Macas, M. MacInnis, D. M. Macleod, S. Márka, Z. Márka, D. V. Martynov, K. Mason, T. J. Massinger, F. Matichard, N. Mavalvala, R. McCarthy, D. E. McClelland, S. McCormick, L. McCuller, J. McIver, T. McRae, G. Mendell, K. Merfeld, E. L. Merilh, F. Meylahn, T. Mistry, R. Mittleman, G. Moreno, C. M. Mow-Lowry, S. Mozzon, A. Mullavey, T. J. N. Nelson, P. Nguyen, L. K. Nuttall, J. Oberling, Richard J. Oram, B. O'Reilly, C. Osthelder, D. J. Ottaway, H. Overmier, J. R. Palamos, W. Parker, E. Payne, A. Pele, R. Penhorwood, C. J. Perez, M. Pirello, H. Radkins, K. E. Ramirez, J. W. Richardson, K. Riles, N. A. Robertson, J. G. Rollins, C. L. Romel, J. H. Romie, M. P. Ross, K. Ryan, T. Sadecki, E. J. Sanchez, L. E. Sanchez, T. R. Saravanan, R. L. Savage, D. Schaetzel, R. Schnabel, R. M. S. Schofield, E. Schwartz, D. Sellers, T. Shaffer, D. Sigg, B. J. J. Slagmolen, J. R. Smith, S. Soni, B. Sorazu, A. P. Spencer, K. A. Strain, L. Sun,

- M. J. Szczepańczyk, M. Thomas, P. Thomas, K. A. Thorne, K. Toland, C. I. Torrie, G. Traylor, M. Tse, A. L. Urban, G. Vajente, G. Valdes, D. C. Vander-Hyde, P. J. Veitch, K. Venkateswara, G. Venugopalan, A. D. Viets, T. Vo, C. Vorvick, M. Wade, R. L. Ward, J. Warner, B. Weaver, R. Weiss, C. Whittle, B. Willke, C. C. Wipf, L. Xiao, H. Yamamoto, Hang Yu, Haocun Yu, L. Zhang, M. E. Zucker, and J. Zweizig. Sensitivity and performance of the advanced LIGO detectors in the third observing run. *Physical Review D*, 102(6), sep 2020.
- [18] Thomas J. Kane and Robert L. Byer. Monolithic, unidirectional single-mode nd:YAG ring laser. *Optics Letters*, 10(2):65, feb 1985.
- [19] I Freitag, A Tünnermann, and H Welling. Power scaling of diode-pumped monolithic nd:YAG lasers to output powers of several watts. *Optics Communications*, 115(5-6):511–515, apr 1995.
- [20] Michèle Heurs, Volker M. Quetschke, Benno Willke, Karsten Danzmann, and Ingo Freitag. Simultaneously suppressing frequency and intensity noise in a nd:YAG nonplanar ring oscillator by means of the current-lock technique. *Optics Letters*, 29(18):2148, sep 2004.
- [21] J. Lui, R. Savage, P. King, L. Zhang, and S. et al. Appert. aLIGO all-bolted PMC summary. LIGO Document T1700543-v2 <https://dcc.ligo.org/LIGO-T1700543>.
- [22] Patrick Kwee, Frank Seifert, Benno Willke, and Karsten Danzmann. Laser beam quality and pointing measurement with an optical resonator. *Review of Scientific Instruments*, 78(7):073103, jul 2007.

- [23] P. Kwee. *Laser Characterization and Stabilization for Precision Interferometry*. PhD thesis, Max Planck Institute for Gravitational Physics, 2010.
- [24] R. S. Abbott and P. J. King. Diode-pumped nd:YAG laser intensity noise suppression using a current shunt. *Review of Scientific Instruments*, 72(2):1346, 2001.
- [25] B W Barr, K A Strain, and C J Killow. Laser amplitude stabilization for advanced interferometric gravitational wave detectors. *Classical and Quantum Gravity*, 22(20):4279–4283, oct 2005.
- [26] Jameson Rollins, David Ottaway, Michael Zucker, Rainer Weiss, and Richard Abbott. Solid-state laser intensity stabilization at the 10^{-8} level. *Optics Letters*, 29(16):1876, aug 2004.
- [27] I Zawischa, M Brendel, K Danzmann, C Fallnich, M Heurs, S Nagano, V Quetschke, H Welling, and B Willke. The GEO 600 laser system. *Classical and Quantum Gravity*, 19(7):1775–1781, mar 2002.
- [28] F Nocera. LIGO laser intensity noise suppression. *Classical and Quantum Gravity*, 21(5):S481–S485, feb 2004.
- [29] P. Horowitz and W. Hill. *The Art of Electronics*. Cambridge University Pr., 2015.
- [30] P. Kwee. aLIGO PSL ISS inner-loop photodiode. LIGO Document D1001998-v2 <https://dcc.ligo.org/LIGO-D1001998/public>.
- [31] C. Bogan. Intensity Stabilization Servo Module (inner loop). LIGO Document D1001985-v2 <https://dcc.ligo.org/LIGO-D1001985/public>.

- [32] G. Mueller. Pointing Requirements for Advanced LIGO. LIGO Document T0900142-v2 <https://dcc.ligo.org/LIGO-T0900142>.
- [33] Euan Morrison, Brian J. Meers, David I. Robertson, and Henry Ward. Automatic alignment of optical interferometers. *Applied Optics*, 33(22):5041, aug 1994.
- [34] H. Kim, P. King, C. Krämer, P. Kwee, J. Pöld, R. Savage, P. Weßels, and B. Willke. Psl final design. DCC LIGO T0900649-v4, February 2010.
- [35] N. Uehara and K. Ueda. Accurate measurement of the radius of curvature of a concave mirror and the power dependence in a high-finesse fabry-perot interferometer. *Applied Optics*, 34(25):5611, sep 1995.
- [36] V. Frolov. DARM Optical Gain and Losses. <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=42231>.
- [37] L. McCuller. Frequency Dependence of Squeezing Loss and Angle, Power Measurement by Radiation Pressure. <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=52718>.
- [38] L. McCuller. Frequency Dependent Squeezing effects in the LIGO Interferometers. LIGO Document G2000863-v2 <https://dcc.ligo.org/LIGO-G2000863>.
- [39] V. Frolov. Arm Cavity Power, Power Recycling Gain, IFO Reflectivity and Arm Cavity Losses. <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=42639>.
- [40] J. Richardson, M. Tse, and C. Austin. Measurement of SRC Gouy Phase. <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=39779>.

- [41] C. Blair. OMC Scans. <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=37742>.
- [42] C. Blair and V. Frolov. 40W OMC Scans Method and Preliminary Data. <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=45830>.
- [43] M. Kasprzack, A. Effler, A. Mullavey, and M. Tse. HAM6 Work - M12 Move. <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=37291>.
- [44] A. Effler and M. Kasprzack. LLO Interferometer Output Beam Study. LIGO Document T1800089-v2 <https://dcc.ligo.org/LIGO-T1800089>.
- [45] C. Blair, V. Frolov, and A. Effler. SRM CO2 Chop. <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=38689>.
- [46] C. Blair. Annular CO2 1W at Various IFO Power. <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=43065>.
- [47] C. Blair, V. Frolov, and A. Effler. OMC Mode Matching in Single Bounce. <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=38698>.
- [48] A. Effler, V. Frolov, and C. Blair. OMC Scans of Cooling Interferometer. <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=41428>.
- [49] A. Green, A. Effler, P. Dupej, V. Frolov, P. Fulda, and H. Yu. Finesse Squeezing Mode Matching Study for LLO. LIGO Document T1900159-v1 <https://dcc.ligo.org/LIGO-T1900159>.
- [50] A. Effler, P. Dupej, H. Yu, A. Green, and C. Blair. SQZ SRM Bounce 98% Mode Matching. <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=42187>.

- [51] Daniel David Brown and Andreas Freise. Pykat, July 2017. <http://www.gwoptics.org/pykat>.
- [52] C. Blair. PRC Gouy phase measurement. <https://alog.ligo-la.caltech.edu/aLOG/index.php?callRep=49409>.
- [53] Haocun Yu, , L. McCuller, M. Tse, N. Kijbunchoo, L. Barsotti, and N. Mavalvala. Quantum correlations between light and the kilogram-mass mirrors of LIGO. *Nature*, 583(7814):43–47, jul 2020.
- [54] Stefan L. Danilishin and Farid Ya. Khalili. Quantum measurement theory in gravitational-wave detectors. *Living Reviews in Relativity*, 15(1), apr 2012.
- [55] E. Knyazev, S. Danilishin, S. Hild, and F.Ya. Khalili. Speedmeter scheme for gravitational-wave detectors based on EPR quantum entanglement. *Physics Letters A*, 382(33):2219–2225, aug 2018.
- [56] Tobias Eberle, Sebastian Steinlechner, Jöran Bauchrowitz, Vitus Händchen, Henning Vahlbruch, Moritz Mehmet, Helge Müller-Ebhardt, and Roman Schnabel. Quantum enhancement of the zero-area sagnac interferometer topology for gravitational wave detection. *Physical Review Letters*, 104(25), jun 2010.
- [57] H. J. Kimble, Yuri Levin, Andrey B. Matsko, Kip S. Thorne, and Sergey P. Vyatchanin. Conversion of conventional gravitational-wave interferometers into quantum nondemolition interferometers by modifying their input and/or output optics. *Physical Review D*, 65(2), dec 2001.
- [58] T. Zhang, E. Knyazev, S. Steinlechner, F. Y. Khalili, B. W. Barr, A. S. Bell, P. Dupej, J. Briggs, C. Gräf, J. Callaghan, J. S. Hennig, E. A. Houston, S. H.

- Huttner, S. S. Leavey, D. Pascucci, B. Sorazu, A. Spencer, J. Wright, K. A. Strain, S. Hild, and S. L. Danilishin. Quantum noise cancellation in asymmetric speed metres with balanced homodyne readout. *New journal of physics*, 20(10):103040, 2018.
- [59] M S Stefszky, C M Mow-Lowry, S S Y Chua, D A Shaddock, B C Buchler, H Vahlbruch, A Khalaidovski, R Schnabel, P K Lam, and D E McClelland. Balanced homodyne detection of optical quantum states at audio-band frequencies and below. *Classical and Quantum Gravity*, 29(14):145015, jun 2012.
- [60] Daniel Sigg and. Status of the LIGO detectors. *Classical and Quantum Gravity*, 25(11):114041, may 2008.
- [61] Brian J. Meers and Kenneth A. Strain. Modulation, signal, and quantum noise in interferometers. *Physical Review A*, 44(7):4693–4703, oct 1991.
- [62] Sebastian Steinlechner, Bryan W. Barr, Angus S. Bell, Stefan L. Danilishin, Andreas Gläfke, Christian Gräf, Jan-Simon Hennig, E. Alasdair Houston, Sabina H. Huttner, Sean S. Leavey, Daniela Pascucci, Borja Sorazu, Andrew Spencer, Kenneth A. Strain, Jennifer Wright, and Stefan Hild. Local-oscillator noise coupling in balanced homodyne readout for advanced gravitational wave detectors. *Physical Review D*, 92(7), oct 2015.
- [63] D. F. Walls. Squeezed states of light. *Nature*, 306(5939):141–146, nov 1983.
- [64] Carlton M. Caves and Bonny L. Schumaker. New formalism for two-photon quantum optics. i. quadrature phases and squeezed states. *Physical Review A*, 31(5):3068–3092, may 1985.

- [65] Peter Fritschel, Matthew Evans, and Valery Frolov. Balanced homodyne readout for quantum limited gravitational wave detectors. *Optics Express*, 22(4):4224, feb 2014.
- [66] Tobin T Fricke, Nicolás D Smith-Lefebvre, Richard Abbott, Rana Adhikari, Katherine L Dooley, Matthew Evans, Peter Fritschel, Valery V Frolov, Keita Kawabe, Jeffrey S Kissel, Bram J J Slagmolen, and Sam J Waldman. DC readout experiment in enhanced LIGO. *Classical and Quantum Gravity*, 29(6):065005, feb 2012.
- [67] D.E. McClelland, N. Mavalvala, Y. Chen, and R. Schnabel. Advanced interferometry, quantum optics and optomechanics in gravitational wave detectors. *Laser & Photonics Reviews*, 5(5):677–696, mar 2011.
- [68] The LIGO Scientific Collaboration. A gravitational wave observatory operating beyond the quantum shot-noise limit. *Nature Physics*, 7(12):962–965, sep 2011.
- [69] L. McCuller, C. Whittle, D. Ganapathy, K. Komori, M. Tse, A. Fernandez-Galiana, L. Barsotti, P. Fritschel, M. MacInnis, F. Matichard, K. Mason, N. Mavalvala, R. Mittleman, Haocun Yu, M.E. Zucker, and M. Evans. Frequency-dependent squeezing for advanced LIGO. *Physical Review Letters*, 124(17), apr 2020.
- [70] Kiwamu Izumi and Daniel Sigg. Advanced LIGO: length sensing and control in a dual recycled interferometric gravitational wave antenna. *Classical and Quantum Gravity*, 34(1):015001, dec 2016.
- [71] S. Dwyer. O3b noise budget. <https://alog.ligo-wa.caltech.edu/aLOG/index.php?callRep=55755>.

- [72] S. Dwyer. personal communication.
- [73] B. Lantz. Optical Integration of the aLIGO isolation tables. LIGO Document G2001539-v2 <https://dcc.ligo.org/G2001539>.
- [74] T. Zhang. *Advanced techniques for future generation gravitational wave detectors*. PhD thesis, University of Glasgow, 2019.
- [75] Hang Yu Peter Fritschel, Vivishek Sudhir. LIGO DCC Document G1702485-v1.
- [76] P. Fritschel and S. Hild. Requirements for the HAM Relay Triple Suspension (HRTS). LIGO Document T1800413-v1 <https://dcc.ligo.org/LIGO-T1900817>.
- [77] L. Barsotti, L. McCuller, M. Evans, and P. Fritschel. The A+ design curve. LIGO DCC, April 2018. LIGO Document T1800042-v5 <https://dcc.ligo.org/LIGO-T1800042/public>.
- [78] C. Bond, P. Fulda, A. Freise, D. Brown, K. Kokeyama, L. Carbone, and A. Perreca. Finesse Input Files for the L1 Interferometer. LIGO Document T1300901-v5 <https://dcc.ligo.org/LIGO-T1300901>.
- [79] B. Lantz, R. Mittleman, J. Kissel, J. Warner, and A. Pele. Typical Motion of HAM Suspension Points in O2. LIGO Document T1800066-v2 <https://dcc.ligo.org/T1800066>.
- [80] A. Huddart. A+ HRTS Preliminary Design Document. LIGO DCC, February 2019. LIGO Document T1900817-v5 <https://dcc.ligo.org/LIGO-T1900817>.

- [81] L. Barsotti. aLIGO ISC Anti-symmetric Port: Beam Parameters and Optical Layout. LIGO Document T1200410-v2 <https://dcc.ligo.org/LIGO-T1200410/public>.
- [82] Peter Fritschel Jeff Lewis Sam Waldman Koji Arai, Sam Barnum. LIGO-T1000276-v5.
- [83] A. Brookes, L. Barsotti, G. Billingsley, K.A. Strain, and M. Zucker. A+ o5-ham6 design considerations and requirements. LIGO DCC, April 2020. E1900385-v1.
- [84] A. Brookes, P. Fritschel, F. Matichard, L. McCuller, M. Noh, and C. Torrie. Sams design requirements document. LIGO DCC, February 2019. E1800314.
- [85] Evgenia Kochkina. *Stigmatic and Astigmatic Gaussian Beams in Fundamental Mode: Impact of Beam Model Choice on Interferometric Pathlength Signal Estimates*. PhD thesis, Hannover U., 2013.
- [86] J. Briggs, M. Barton, J. Jones, E. Oelker, S. Webster, and K.A. Strain. A+ BHD ISC Additional Design Requirements. LIGO Document E2000072-v2 <https://dcc.ligo.org/LIGO-E2000072-v2>.
- [87] Ryan Lawrence, Michael Zucker, Peter Fritschel, Phil Marfuta, and David Shoemaker. Adaptive thermal compensation of test masses in advanced LIGO. *Classical and Quantum Gravity*, 19(7):1803–1812, mar 2002.
- [88] C. Blair. personal communication.

- [89] P. Fulda, R. Goetz, A. Green, R. Martin, and D. B. Tanner. A+ Output Faraday Isolator Final Design. LIGO Document T1900788-v2 <https://dcc.ligo.org/LIGO-T1900788>.
- [90] F. Bayer-Helms. Coupling coefficients of an incident wave and the modes of a spherical optical resonator in the case of mismatching and misalignment. *Applied Optics*, 23(9):1369, may 1984.
- [91] Antonio Perreca, Aidan F. Brooks, Jonathan W. Richardson, Daniel Töyrä, and Rory Smith. Analysis and visualization of the output mode-matching requirements for squeezing in advanced LIGO and future gravitational wave detectors. *Physical Review D*, 101(10), may 2020.
- [92] A.E. Siegman. *lasers*. University Science Books, 1986.
- [93] A. A. Muzammil, C. Mueller, and G. Mueller. Optical Layout and Parameters for the Advanced LIGO Cavities. LIGO Document T0900043-v11 <https://dcc.ligo.org/LIGO-T0900043/public>.
- [94] LIGO Optics Database. <https://galaxy.ligo.caltech.edu/optics/>.
- [95] R. Martin and G. Billingsley. Measurement Uncertainties in Determining the Radius of Curvature for SR2-04. LIGO Document T1300653-v1 <https://dcc.ligo.org/LIGO-T1300653>.
- [96] G. Billingsley. SR301 Figure Measurement. LIGO Document E1101231-v1 <https://dcc.ligo.org/E1101231-v1>.
- [97] P. Fritschel and E. Gustafson. Advanced LIGO Optical Layout. LIGO Document D0902838-v5 <https://dcc.ligo.org/LIGO-D0902838>.

- [98] M. Smith and D. Coyne. Advanced LIGO L1 Optical Layout, ZEMAX. LIGO Document D0902216-v8 <https://dcc.ligo.org/LIGO-D0902216>.